AIRLINE SENTIMENT ANALYSIS

Business Value Document

Prepared for Executive Leadership | DistilBERT NLP Model v13 | December 2025

01 - The Bottom Line - Financial & Operational Summary

Value Proposition

Converting 14,604 daily tweets from unstructured digital noise into real-time operational intelligence - at 82.83% accuracy and 0.12s response time - eliminating 8,000+ hours of annual manual complaint classification.

Investment vs. Return - What We Spent vs. What We Save

~$18K	>$240K	~1,200%
Initial Investment Development cost + Infrastructure	Projected Annual Savings Replacing 8,000+ hours of manual work	Expected ROI Within the first year

82.83%	0.12 sec	14,604
Classification Accuracy Exceeded the 75% target threshold	Inference Speed Optimized for real-time monitoring	Tweets Analyzed Across 6 major airlines

Project Status - Current Snapshot

Status	Green Light - Model promoted to Production (v13) via MLflow
Goals	All technical benchmarks met and exceeded
Timeline	Delivered on schedule - within a 4-week development sprint
Current Phase	Phase 1 complete - ready to enter Phase 2

02 - Strategic Context - Why This Project, Why Now?

Market Need - The Problem We're Solving

Airlines receive thousands of social media mentions every day. 62.72% of those are negative, and 4 out of 5 complaints are about human service interactions - not operations. The problem is not the aircraft; it's the frontline staff and escalation protocols.

Without the Solution	With the Solution
Teams manually classify every complaint	Instant automated classification - zero manual effort
Responses delayed by hours or days	Real-time alerts for critical complaints
Negative content spreads unchecked	Live management monitoring dashboard
Management lacks competitor benchmarking data	Visual airline performance benchmarking
US Airways: 77.7% negative - zero automated intervention	*Targeted response within under one hour*

Strategic Alignment - How This Serves Company Goals

Goal	Business Impact
Employee Efficiency	Frees monitoring teams from manual classification so they focus on response and resolution
Data-Driven Decisions	Management receives live reports instead of weekly manual summaries
Competitive Advantage	Southwest leads with 23.6% positive sentiment - we now have a measurable roadmap to match it
Response Speed	Automated alert system ensures critical complaints are addressed within one hour

03 - Impact & Outcomes - Real-World Results

Accuracy in Business Terms

The model classifies 8 out of every 10 tweets correctly. Critical complaints (negative sentiment) are detected at 90% accuracy - meaning 90 out of every 100 angry tweets are automatically flagged for immediate action.

90%	66%	78%
Negative Complaints Detection accuracy for critical complaints	Neutral Tweets Improved - mixed language and tone	Positive Tweets Significant improvement in Version 13

Efficiency Gains - Time & Effort Savings

Task	Without Tool	With Tool
Classifying 14,604 tweets	~730 hours	< 30 minutes
Detecting a critical complaint	4-24 hours	< 0.12 seconds
Airline comparison report	1 full week	Instant - Live Dashboard
Complaint pattern analysis	Not available	Automated + Categorized

Key Business Findings

Human service is the #1 complaint driver: 31.7% of all negative tweets - not flight delays or lost baggage
United Airlines faces a reputation crisis: 68.9% negative sentiment - requires urgent intervention
US Airways is the worst performer: 77.7% negative - the highest rate in the sector
Southwest is the benchmark to follow: 23.6% positive - a replicable blueprint for improvement
Peak complaint window is 6-9 AM: Volume surges from 131 tweets at midnight to 999 by 9 AM
Negative content spreads faster: Retweet rate 0.093 vs. 0.070 for positive - a real viral risk

04 - Project Roadmap & Milestones - Timeline & Delivery

Phase	Timeline	Activity	Status
Phase 0	Completed (Dec 2025)	Model development - Fine-tune DistilBERT v13 and promote to Production via MLflow	Done
Phase 1	Month 1 (Jan 2026)	Deploy FastAPI + Streamlit + configure automated alerts for United and US Airways accounts	In Progress
Phase 2	Months 1-3 (Q1 2026)	CRM integration - route high-severity complaints directly to Service Recovery teams	Scheduled
Phase 3	Months 3-6 (Q2 2026)	Expand to Instagram and Facebook + multilingual model deployment	Planned

Go-to-Market Date: The tools will be available to monitoring teams within the first month of Phase 1 approval. Customer service teams will begin receiving intelligent alerts from Day 1.

05 - Risks & Mitigation - We Have a Plan B

Risk	Potential Impact	Mitigation Plan
Evolving tweet language & context drift	Accuracy degrades over time without updates	Plan: Quarterly retraining with 5,000+ fresh tweets
High GPU server costs	Real-time inference halted if budget is exceeded	Alternative: AWS/GCP Spot Instances - up to 70% cost reduction
Neutral class accuracy gap (66%)	Errors on ambiguous mixed-sentiment tweets	Plan: Build ensemble model in Phase 3 (BERT + LSTM)
Geographic data bias (Boston)	33.4% of data from Boston - regional skew	Plan: Expand dataset with national coverage in Phase 2

06 - The Call to Action - What We Need From You

We need your decisions on three fronts to move from Phase 1 into Phase 2 and scale fully:

#	Decision Required	Details	Priority
1	Approve GPU server budget for production	~$800-1,200/month (AWS/GCP) to sustain 0.12s inference in the production environment	Urgent
2	Authorize hiring 4-6 social monitoring agents	Shift: 6 AM - 12 PM (peak hours) to act on automated alerts in real time	Urgent
3	Provide direction on rolling out the tool to other departments	Should we expand to internal analytics - employee reviews, flight satisfaction surveys?	Medium

Airlines that invest in frontline staff empowerment and proactive communication will differentiate themselves in a market where customer expectations are high but satisfaction is consistently low. Southwest Airlines provides a replicable blueprint - and this system gives every airline the data infrastructure needed to follow it.